
AN EXAMPLE PAPER FOR THE PAPER TEMPLATE

PREPRINT

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ABSTRACT

This is the worked example shipped with the paper template. It exists to prove that one command fetches experiment results from the tracker, regenerates every figure and table from them, and compiles this document—so that a number in a paper always has a traceable path back to the runs that produced it. Replace this paper with your own; the wiring is the point.

Keywords reproducibility, research infrastructure, LaTeX

1 Introduction

Experiment code and the paper written about it normally live in separate repositories, and the numbers travel between them by hand. They then go stale silently: nothing connects the figure in the PDF to the runs it summarizes, and a table copied out of a notebook keeps its authority long after the sweep that produced it was rerun. This repository closes that gap without merging the two codebases: the paper fetches its numbers from the experiment tracker, records which runs it fetched, and fails to build when the two disagree.

Contributions

- C1.** A committed, human-readable cache of experiment-tracker results, with the exact run identifiers behind it pinned in a lockfile, so a paper's numbers are reproducible without access to the tracker.
- C2.** Figures and tables generated from that cache with recorded provenance, which makes staleness a build failure rather than a reading error.
- C3.** A document directory that compiles standalone—on Overleaf, and as a flattened submission bundle—and whose prose survives a change of venue template.

2 Method

The synthetic study behind fig. 2 stands in for a real experiment. Deep reinforcement learning from pixels [Mnih et al., 2015] is sample-inefficient, which motivated episodic control methods [Blundell et al., 2016, Pritzel et al., 2017]. Oquab et al. [2024] showed that self-supervised features transfer broadly; replacing an episodic controller's state encoder with such features is the kind of question this repository layout is built to support.

Hand-authored diagrams live beside the data-figure configs, as `.drawio` sources. Figure 1 is one: it is drawn by hand, not fetched from the tracker, and its PDF is a generated export.

3 Results

Neither fig. 2 nor table 1 is drawn or typed by hand. Both are rendered from `data/learning_curves.csv` and `data/final_scores.csv`, which make `fetch` writes from the runs pinned in `data/wandb.lock.yml`. Each output records the content hash of the file it read, so changing the data without rebuilding fails `make check-generated` instead of quietly shipping a stale result.

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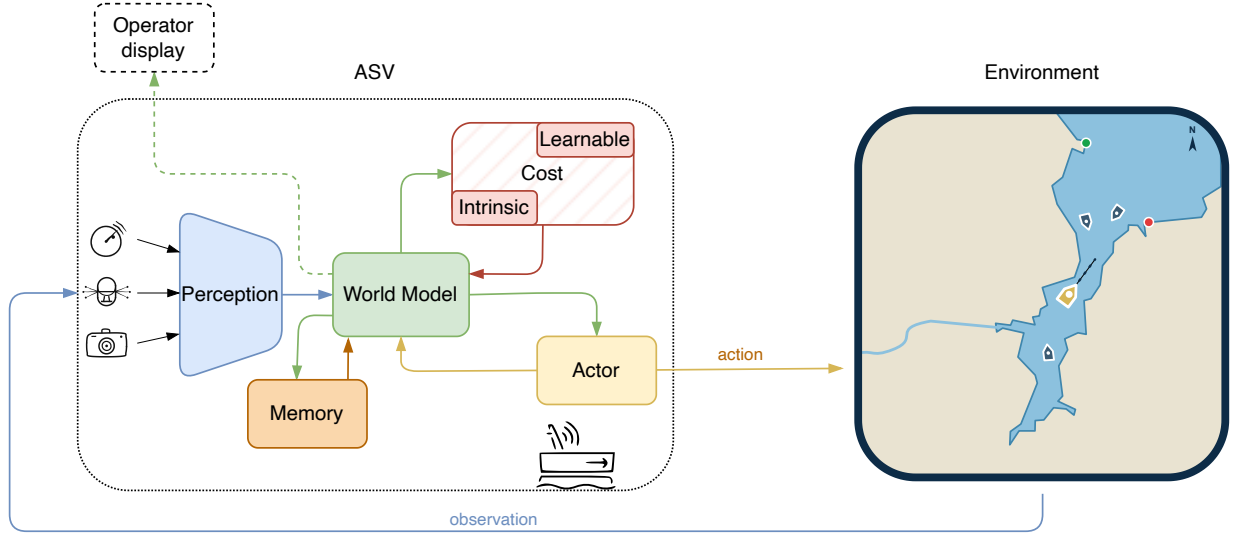


Figure 1: Example schematic: a world-model control loop. The source is `figures/architecture.drawio`; regenerate the export with `make drawio`.

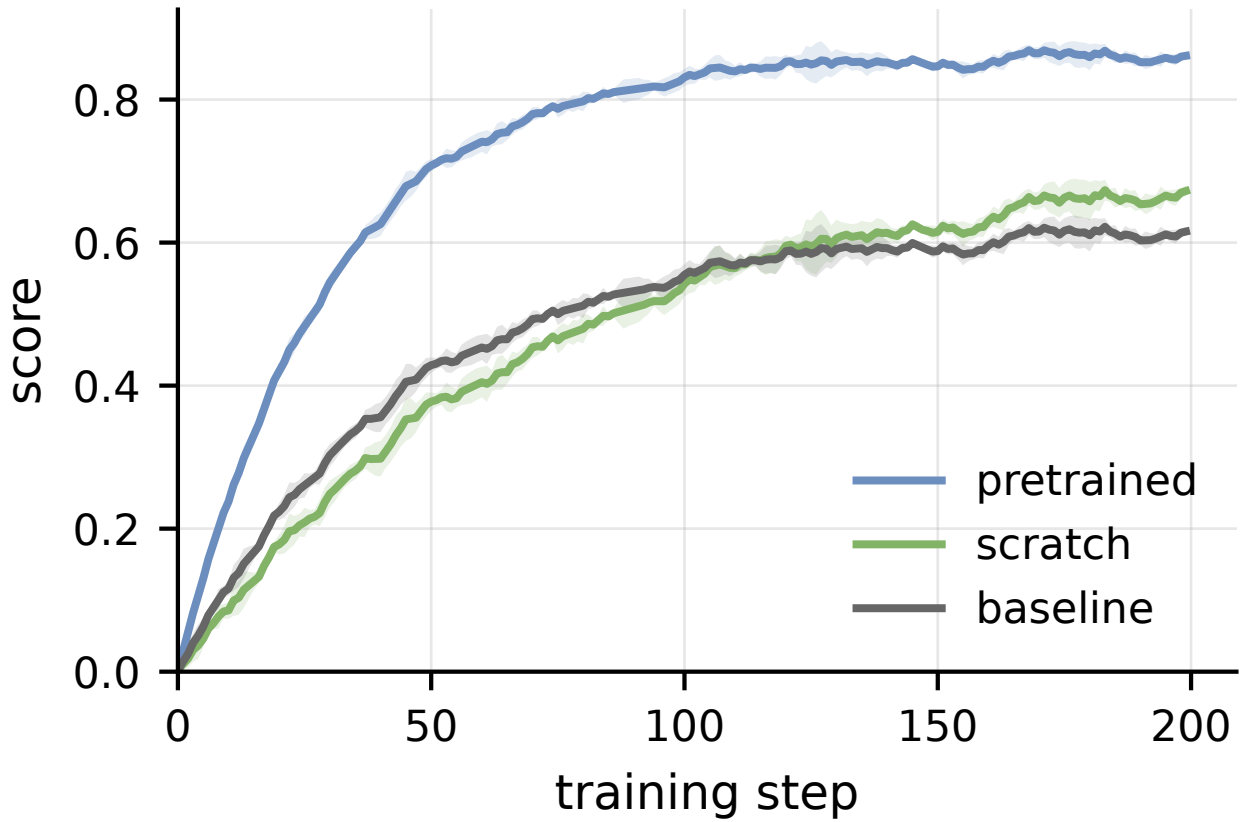


Figure 2: Learning curves for the synthetic example study: mean over five seeds, shaded band is one standard deviation. Regenerate with `make plots`.

Table 1: Final score after training, mean over five seeds. The tabular is generated; the caption, label and float placement are not.

Method	Score	SD	Seeds
pretrained	0.86	0.01	5
scratch	0.67	0.01	5
baseline	0.62	0.01	5

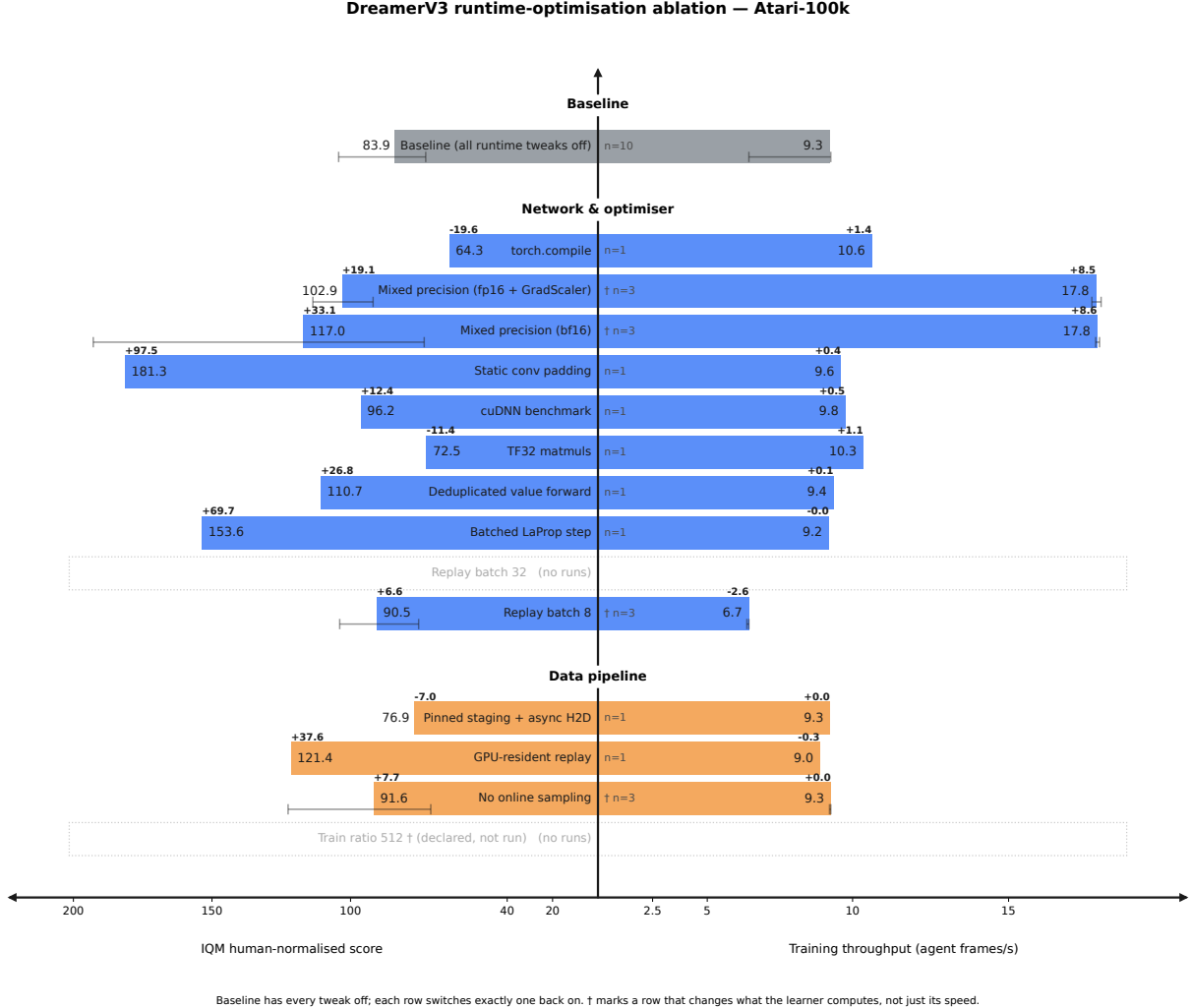


Figure 3: DreamerV3 runtime-optimisation ablation on Atari-100k. Rendered from data/figures/dreamer_optimisations_ablation.csv, pinned to the runs in data/wandb.lock.yml. Regenerate with make plots.

4 Conclusion

Delete this paper and write your own. Keep the structure: one bibliography and one cache as sources of truth, generated files committed where the compiler needs them, and a check that tells you when the two have drifted apart.

References

Volodymyr Mnih, Koray Kavukcuoglu, David Silver, Andrei A. Rusu, Joel Veness, Marc G. Bellemare, Alex Graves, Martin Riedmiller, Andreas K. Fidjeland, Georg Ostrovski, Stig Petersen, Charles Beattie, Amir Sadik, Ioannis

- Antonoglou, Helen King, Dharshan Kumaran, Daan Wierstra, Shane Legg, and Demis Hassabis. Human-level control through deep reinforcement learning. *Nature*, 518(7540):529–533, 2015. doi:10.1038/nature14236.
- Charles Blundell, Benigno Uria, Alexander Pritzel, Yazhe Li, Avraham Ruderman, Joel Z. Leibo, Jack Rae, Daan Wierstra, and Demis Hassabis. Model-free episodic control. In *arXiv preprint arXiv:1606.04460*, 2016.
- Alexander Pritzel, Benigno Uria, Sriram Srinivasan, Adrià Puigdomènech Badia, Oriol Vinyals, Demis Hassabis, Daan Wierstra, and Charles Blundell. Neural episodic control. In *Proceedings of the 34th International Conference on Machine Learning (ICML)*, pages 2827–2836, 2017.
- Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy V. Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al. DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*, 2024.